

Application Scenarios

Light yellow powder | Good water solubility | Clear solution | Pleasant taste | Low GI | 0 sugar 0 fat
Recommended Daily Intake: ≤4g/day
Suitable for formula development and application of terminal products such as compressed candies, solid beverages, functional drinks, dairy products, energy bars, etc.

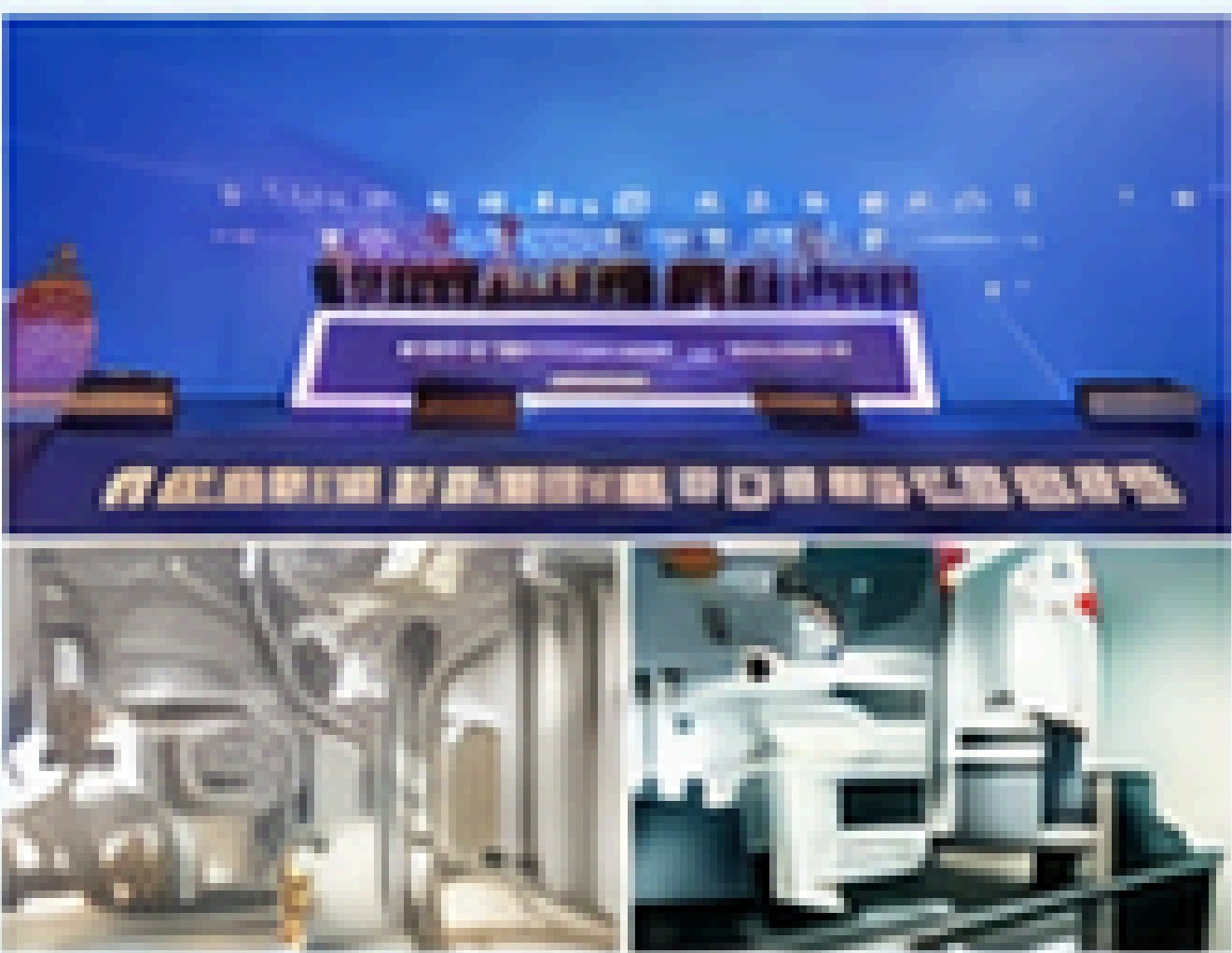
Warm Tips: Not suitable for infants, pregnant and lactating women



About the Manufacturer

Qingdao Dakang Marine Biotechnology Co., Ltd. is a joint venture established by Qingdao Gather Great Ocean Algae Industry Group Co., Ltd. and Qingdao Marine Biomedical Research Institute Co., Ltd. It is a modern biomedical enterprise focusing on the R&D, production, and high-value application of marine functional oligosaccharides. Relying on the core technological achievements of Academician Guan Huashi's team from the Chinese Academy of Engineering - "Preparation Technology of Marine Characteristic Sugars (Sugar Library Construction) and Application Development" (First Prize of National Technological Invention Award), the company is committed to becoming a world-renowned professional manufacturer of marine functional oligosaccharides and a source of high-end marine oligosaccharide raw materials.

As the industrialization base of Qingdao International Oligosaccharide Preparation Center, the company adheres to the concept of "Technology Empowerment, Breaking Barriers", sequentially arranges the R&D of a series of national new food raw materials, and opens up the "last mile" of scientific research achievement transformation. On February 5, 2026, the alginate oligosaccharide declared by the company was officially approved as a national new food raw material, achieving a "zero breakthrough" in the application of alginate oligosaccharide in the food field. In the future, building on its core technology for preparing marine characteristic oligosaccharides, the company will lead the innovative development of the new-generation marine oligosaccharide prebiotic industry and provide global customers with high-quality, high-performance oligosaccharide raw materials and innovative solutions.



Canadian Distributor: Golden Ocean Industrial Co., Ltd.
Address: Unit 140, 8877 Odlin Crescent, Richmond, BC V6X 3Z7
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Alginate Oligosaccharide New Food Raw Material

Wei Shi Xin Tong Zi [2026] No.0004

NEW GENERATION MARINE FUNCTIONAL FACTOR, PROTECTING GUT-BRAIN HEALTH

Exclusive Qualification

The first domestic manufacturer approved for alginate oligosaccharide as a new food raw material



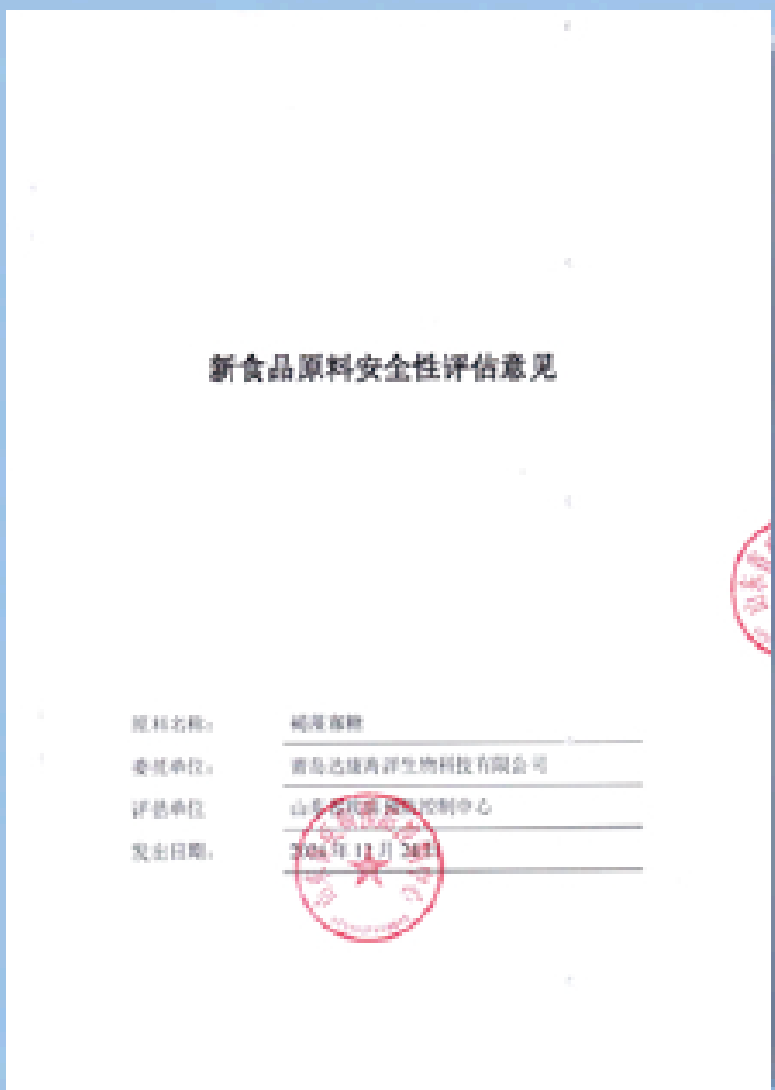
Technological Leadership

Relying on the marine characteristic oligosaccharide preparation technology of Academician Guan Huashi's team (First Prize of National Technological Invention Award)



Safety Certification

First to complete systematic safety research and certified by authoritative institutions



Scientific Verification

New generation marine prebiotics targeting second-generation probiotics



Golden Ocean Industrial Co., Ltd.

Canadian Distributor of Alginate Oligosaccharide

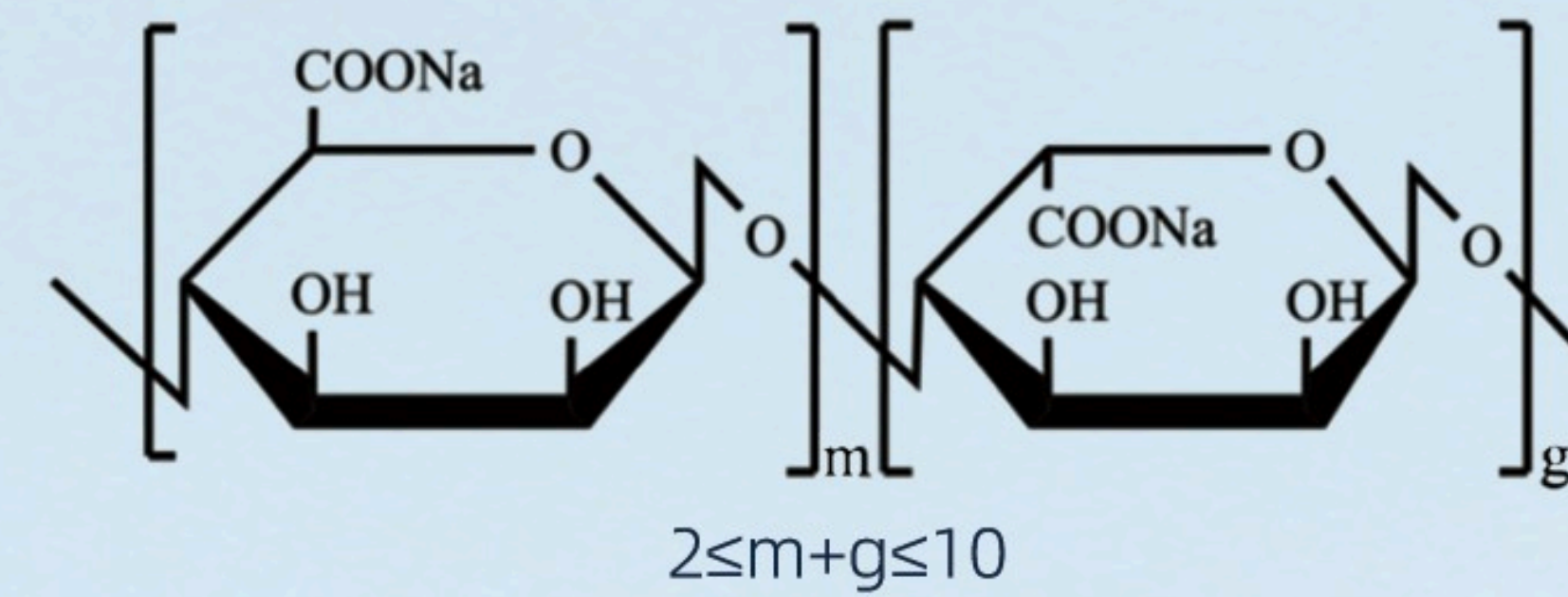


Product Introduction

Alginate Oligosaccharide (AOS), also known as algin oligosaccharide, is a saturated oligosaccharide with a polymerization degree of 2~10 formed by the alternate connection of sodium β -D-mannuronate and sodium α -L-guluronate through 1,4-glycosidic bond

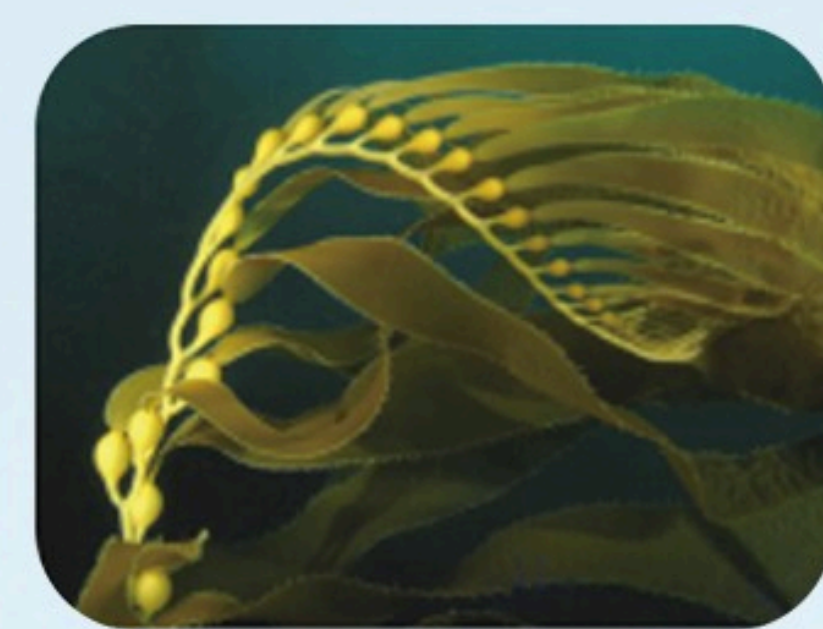
Molecular Formula: $(C_6H_7O_6Na)_n, n=2 \sim 10$.

Structural Formula:



Preparation Introduction

The alginate oligosaccharide produced by Qingdao Dakang Marine Biotechnology Co., Ltd. is made from sodium alginate through processes such as hydrochloric acid dissolution, high-temperature degradation, neutralization, decolorization, purification, concentration and drying.



Brown Algae



Sodium Alginate



Degrade, Purify and Dry



Alginate oligosaccharides

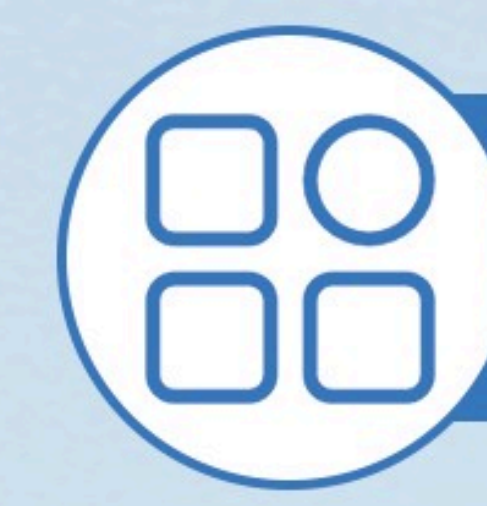


Technical Indicators

Sensory Requirements	
Items	Requirement
Sensory Requirements	With inherent taste and odor of this product, no peculiar smel

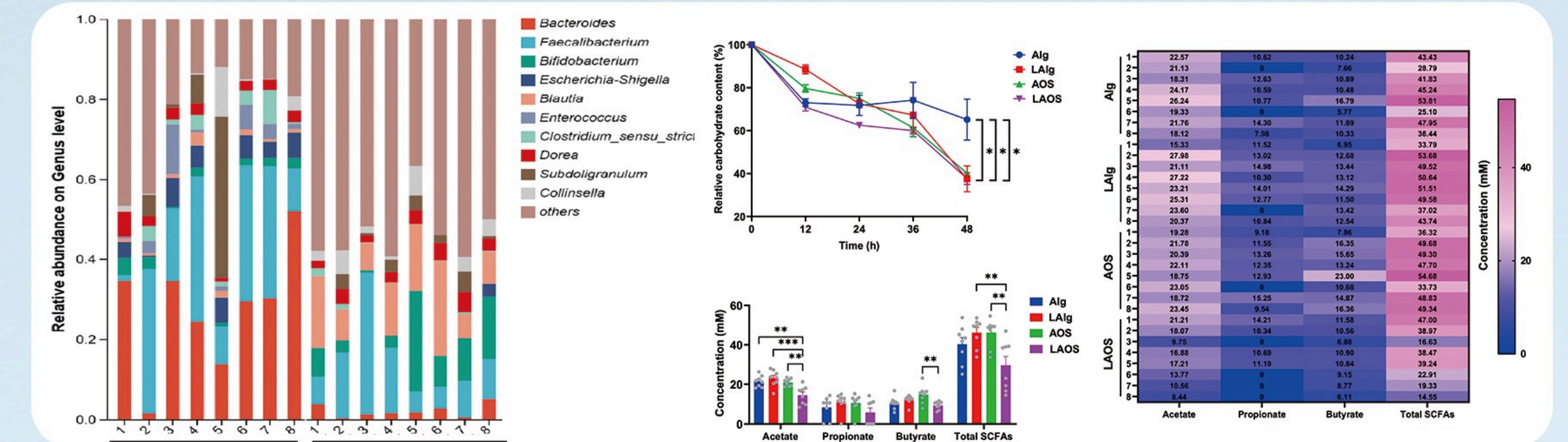
Microbiological Indicators	
Items	Indicator
Total Plate Count, CFU/g	≤ 1000
Escherichia coli, CFU/g	≤ 10
Molds and Yeasts, CFU/g	≤ 50
Staphylococcus aureus, per 25 g	Not Detected
Salmonella, per 25 g	Not Detected

Physical and Chemical Indicators	
Items	Indicator
Content of Brown Algae Oligosaccharides (calculated as sodium uronate, on dry basis), %	≥ 90.0
Average Molecular Weight, Da	600 ~ 1400
Average Degree of Polymerization	3.0 ~ 7.0
Moisture, g/100g	≤ 10.0
Ash, g/100g	20.0 ~ 27.0
pH Value (10 g/L aqueous solution)	5.5 ~ 6.5
Lead (Pb), mg/kg	≤ 2.0
Total Arsenic (As), mg/kg	≤ 1.0
Total Mercury (Hg), mg/kg	≤ 0.3



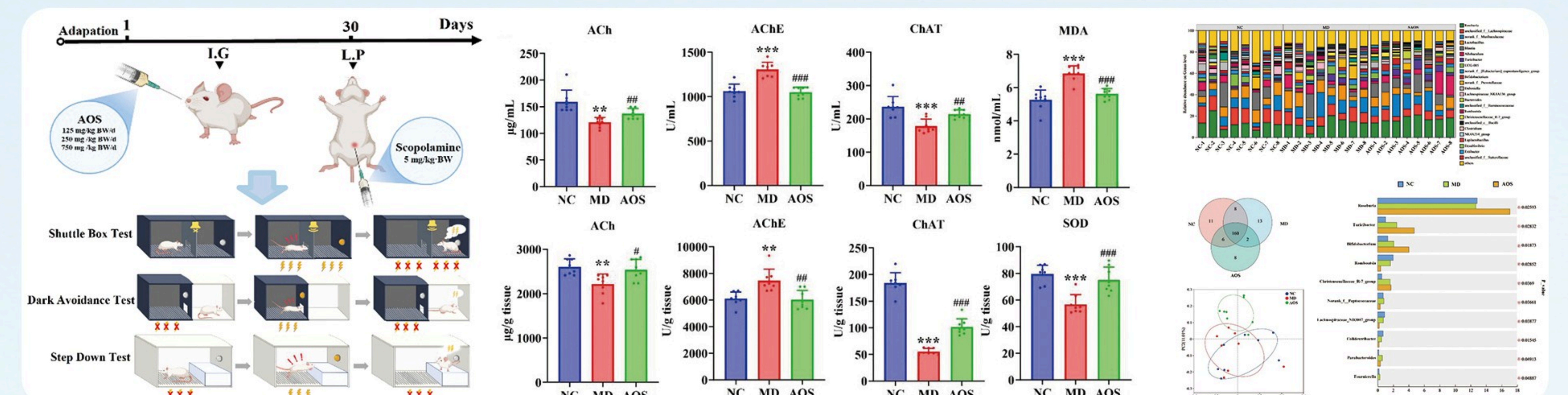
Nutrition and Functions

1. Prebiotic. Verified by animal experiments, as a new generation prebiotic, alginate oligosaccharide can significantly increase the abundance of second-generation probiotics such as *B.xylanisolvans* and *F.prausnitzii*, promote the synthesis of short-chain fatty acids, and improve intestinal immunity.



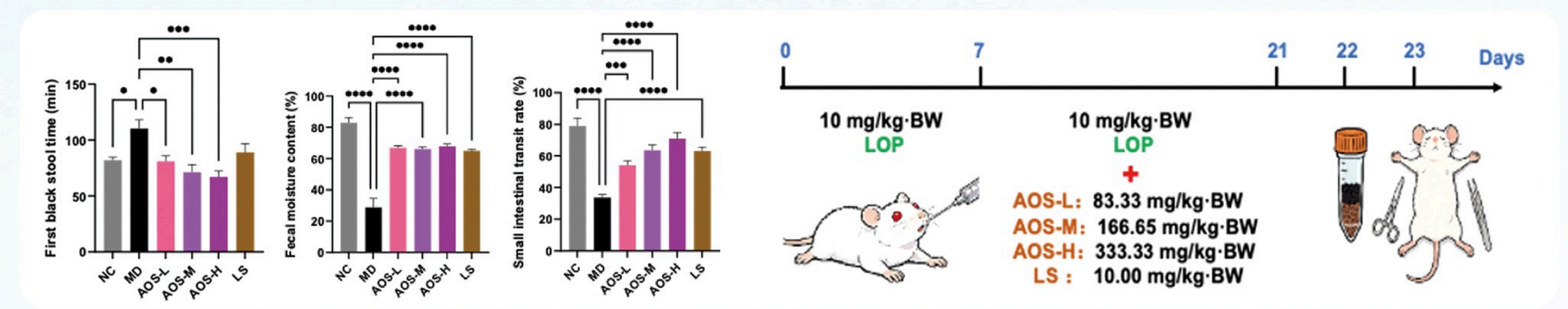
Effects of Alginate Oligosaccharides on Gut Microbiota and Its Secondary Metabolite Short-Chain Fatty Acids

2. Memory Improvement. Verified by animal experiments, alginate oligosaccharide has a significant memory improvement effect on normal aging animals and animals with memory impairment induced by scopolamine. It can intervene in the endocannabinoid system and cholinergic system by regulating intestinal flora, and exert cognitive improvement effect through a synergistic network.



Algin oligosaccharides can significantly improve memory in both normally aging animals and animals with induced memory impairment

3. Moistening Intestines and Relieving Constipation. Verified by animal experiments, alginate oligosaccharide can significantly shorten the first defecation time of experimental animals, improve small intestinal propulsion rate, increase fecal water content, restore the integrity of intestinal wall villi, and has a significant effect of moistening intestines and relieving constipation.



Moistening Intestine and Laxative Effect of Alginate Oligosaccharides on Constipated Mice